`bell`, `note`, `heart`, `fish`, `lock`, and `unlock`. We then made an array called `characters[]` to create an index of each of these individual character names. Because this is an array of arrays, similar to our discussion of character arrays in the last chapter, we need to use the "*" symbol after the data type to indicate that this is an array of pointers—basically references to places in memory where the arrays are stored.

In our `for` loop, we created each character in turn using the following line:

```
lcd.createChar(i, ((byte*)characters[i]));
```

Here, our character number is determined by the `for` loop counter and the name of the array is determined by the index of the `characters[]` array. To get this array to point to the correct character array, we need to remind the compiler that `characters[i]` is a pointer by adding `(byte*)` in front of the array name bracketed by parenthesis. This turns that array into a pointer that the compiler can reference, creating the appropriate character. I know I said we wouldn't talk much about pointers, but it's not that hard and plenty of reference material abounds if you want find out more.

Example Code: Fish Tank Animation

We are almost done with the LiquidCrystal library, with only one more example and a couple new functions to go. By using these new functions with some custom characters, we can create a rather low-tech 8-bit animation—maybe something like a fish swimming in a fish tank. The example in Listing 9-3 takes two strategies for creating an animation. First, it moves the character around the screen by using functions that scroll the data to the left or right. Second, it displays multiple characters at the same location with a delay in between each one to create a simple flipbook style animation.

To make this code work, we designed multiple fish characters that include different directions the fish might swim, as well as a few characters so that the fish can blow bubbles. Four functions are created to control the fish's movement: `scrollLeft()`, `scrollRight()`, `stopCenter()`, and `blowBubbles()`. By calling each function, and passing the number of steps in the case of the scrolling functions, we can get the fish to move around the LCD while keeping track of it the entire time using the variables x and y.

Listing 9-3. *Fish Tank Animation Source Code*

```
#include <LiquidCrystal.h>
LiquidCrystal lcd(5, 6, 7, 8, 9, 10);

byte fishLeft[8] = {B00000,B00000,B00000,B00000,B01101,B11110,B01101,B00000};
byte fishRight[8] = {B00000,B00000,B00000,B00000,B10110,B01111,B10110,B00000};
byte fishCenter[8] = {B00000,B00000,B00000,B00000,B00100,B01110,B00100,B00000};
byte fishBubbles1[8] = {B00010,B00000,B00100,B00010,B00100,B01110,B00100,B00000};
byte fishBubbles2[8] = {B00000,B00100,B00010,B00000,B00100,B01110,B00100,B00000};
byte fishBubbles3[8] = {B00100,B00000,B00000,B00000,B00100,B01110,B00100,B00000};

byte x = 0;
byte y = 0;

int time = 600;

void setup() {
  lcd.begin(16,2);
  lcd.createChar(0, fishBubbles1);
  lcd.createChar(1, fishBubbles2);
  lcd.createChar(2, fishBubbles3);
```